

I briefly met with Ashley, Layton, and Joseph to discuss the homework on Thursday 9/10, so some of these ideas came as a result of our collaboration.

Lemma 1. *Any closed subset of a compact space is compact.*

Proof. Consider any open cover $\{U_\alpha\}$ of C . Since C is closed, $K \setminus C$ is open. So then $\{U_\alpha\} \cup \{K \setminus C\}$ must be an open cover for all of K . Since K is compact, there exists some finite subcover of $\{U_\alpha\} \cup \{K \setminus C\}$ which covers all of K . Well, $C \subseteq K$, so this finite subcover must cover C as well.

Note that $K \setminus C$ cannot cover $C \subseteq K$ and therefore can't cover all of K . So some subset of $\{U_\alpha\}$ must be included in this subcover. Let $\{U_{\alpha_i}\}_{i=1}^n$ be this subset. If $X \setminus C$ is in this subcover, notice that $X \setminus C$ is disjoint from C , and so $\{U_{\alpha_i}\}_{i=1}^n$ must cover C . Otherwise we once again have that $\{U_{\alpha_i}\}_{i=1}^n$ covers K and therefore C .

So any open cover $\{U_\alpha\}$ of C has a finite subcover, and therefore C is compact. □

Lemma 2. *The continuous image of a compact space is compact.*

Proof. Let K be compact and $f : K \rightarrow Y$ a continuous map. Now consider any open cover $\{U_\alpha\}$ of $f(K)$. Since f is continuous, each preimage $f^{-1}(U_\alpha)$ is open in K . Now, $\forall x \in K, f(x) \in U_\alpha$ for some α . Therefore, $x \in f^{-1}(U_\alpha)$, and we see that $\{f^{-1}(U_\alpha)\}$ is an open cover for K . Then since K is compact, there exists a finite subcover $\{f^{-1}(U_{\alpha_i})\}_{i=1}^n \subseteq \{f^{-1}(U_\alpha)\}$.

Finally, $K \subseteq \bigcup_{i=1}^n f^{-1}(U_{\alpha_i}) \implies f(K) \subseteq \bigcup_{i=1}^n f(f^{-1}(U_{\alpha_i})) = \bigcup_{i=1}^n U_{\alpha_i} \implies \{U_{\alpha_i}\}_{i=1}^n \subseteq \{U_\alpha\}$ is a finite subcover. So any open cover $\{U_\alpha\}$ of $f(K)$ has a finite subcover, and therefore $f(K)$ is compact. □

Lemma 3. *Any compact subset of a Hausdorff space is closed.*

Proof. Let H be Hausdorff and $K \subseteq H$ a compact subset. We show that $H \setminus K$ is open.

Since H is Hausdorff, for each $x \in H \setminus K$ and each $k \in K$, we can find disjoint open sets $U_k(x), V_k$ containing x and k , respectively. Now, $\{V_k \mid k \in K\}$ is an open cover of K , and since K is compact there exists a finite subcover $\{V_{k_i}\}_{i=1}^n \subseteq V_k$. Set $U_x = \bigcap_{i=1}^n U_{k_i}(x)$ and recall that $U_{k_i}(x) \ni x$ is disjoint with V_{k_i} for each $i \in [n]$.

So by construction, U_x is disjoint from $\bigcup_{i=1}^n V_{k_i} \supseteq K$ placing it completely inside $H \setminus K$. That is, $U_x \subseteq H \setminus K, \forall x \in H \setminus K$. On the otherhand, $x \in U_x, \forall x \in H \setminus K \implies H \setminus K \subseteq \bigcup_{x \in H \setminus K} U_x$.

Finally, we see that then $H \setminus K = \bigcup_{x \in H \setminus K} U_x$, an arbitrary union of open sets, which implies that $H \setminus K$. Therefore, it's complement K must be closed. □

We used *Heine-Borel* several times in proofs during the first lecture, so I use it here.

Lemma 4. S^n is compact for all $n \in \mathbb{N}$.

Proof. Take S^n to be the unit n -sphere in $\mathbb{R}^{n+1}$. In other words, we define $S^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}$ as a subspace of $\mathbb{R}^{n+1}$. Clearly it is bounded since it fits in the open ball $B_\pi(\mathbf{0})$. Now consider $\mathbb{R}^{n+1} \setminus S^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| < 1\} \cup \{x \in \mathbb{R}^{n+1} \mid \|x\| > 1\}$. $B_1(\mathbf{0})$ is the first partite set. Then $\bigcup_{\|x\| > 1} B_{\frac{1-\|x\|}{2}}(x)$ is the second, since it includes an open ball containing every point 'outside' of S^n , and the radius of these balls is half the distance from x to S^n . All together, we get that $\mathbb{R}^{n+1} \setminus S^n$ is an arbitrary union of open balls in $\mathbb{R}^{n+1}$, and therefore open. So S^n is closed and bounded, and therefore by *Heine-Borel* it is compact. $\square$

Theorem 5. If $f : X \hookrightarrow Y$ is a continuous injection where X is compact and Y is Hausdorff, then $X \cong f(X)$. If f is surjective then f is a homeomorphism.

Proof. Since f is injective, X is in bijection with its image $f(X)$. So we show $f^{-1} : f(X) \rightarrow X$ is continuous by showing the preimage of every closed set is closed.

Consider any closed subset $C \subseteq X$. Since X is compact, by **Lemma 1** we have that C is also compact. Then, since f is continuous, by **Lemma 2**, the continuous image (and preimage of C via f^{-1}) $f(C)$ of the compact set C is also compact. Finally, by **Lemma 3**, any compact subset of a Hausdorff space is closed. So $f(C) \subseteq f(X) \subseteq Y \implies f(C)$ is closed. So we see the preimage $(f^{-1})^{-1}(C) = f(C)$ of any closed set via f^{-1} is closed, and so f^{-1} is also continuous.

So f is a bicontinuous bijection from X to $f(X)$, or equivalently, f is a homeomorphism from X to its image and $X \cong f(X)$. Finally, if $f(X) = Y$ of course f is a homeomorphism from X to Y . $\square$

Theorem 6. $B^{2n}/S^{2n-1} \cong S^{2n}$

Proof. Let $\pi : B^m \rightarrow B^m/S^{m-1}$ be the quotient map which collapses the boundary S^{m-1} of the ball to a single point. Define $h : B^m \rightarrow S^m$ by

$$h(x) = \left(2x\sqrt{1-\|x\|^2}, 1-2\|x\|^2 \right).$$

Note that $x \in S^m \iff |x| = 1 \iff |x|^2 = 1$. So this map indeed goes into the codomain since $|(2x\sqrt{1-|x|^2}, 1-2|x|^2)|^2 = (2x\sqrt{1-|x|^2})^2 + (1-2|x|^2)^2 = 4|x|^2(1-|x|^2) + 1 - 4|x|^2 + 4|x|^4 = 1$.

If $x \in S^{m-1}$, then $\|x\| = 1$, and therefore $h(x) = (2x\sqrt{1-|x|^2}, 1-2|x|^2) = (\mathbf{0}, -1)$. (*)

So h induces a useful map to $H : B^m/S^{m-1} \rightarrow S^m$ defined by $H([x]_{S^{m-1}}) = h(x)$. If $[x]_{S^{m-1}} = [y]_{S^{m-1}}$ then $x, y \in S^{m-1}$ or $x, y \in B^{2m} \setminus S^{m-1}$. If the former holds, then $h(x) = h(y) = (\mathbf{0}, -1)$ by (*). Otherwise if the latter holds by definition $x = y \implies h(x) = h(y)$. so in either case, $H([x]_{S^{m-1}}) = H([y]_{S^{m-1}}) = h(x) = h(y)$. So H is well-defined.

Now, suppose $H([x]_{S^{2m-1}}) = H([y]_{S^{2m-1}}) = h(x) = h(y)$. If $x, y \in S^{2m-1}$, then of course $[x]_{S^{2m-1}} = [y]_{S^{2m-1}}$ since all of S^{2m-1} is collapsed to a single point. Otherwise $x, y \in B^{2m} \setminus S^{m-1} \implies$

$$(2x\sqrt{1-|x|^2}, 1-2|x|^2) = (2y\sqrt{1-|y|^2}, 1-2|y|^2) \implies \begin{cases} 2x\sqrt{1-|x|^2} = 2y\sqrt{1-|y|^2} \\ 1-2|x|^2 = 1-2|y|^2 \end{cases} \quad \text{and then}$$

the second row gives $|x| = |y|$ and so $2x\sqrt{1-|x|^2} = 2y\sqrt{1-|y|^2} = 2y\sqrt{1-|x|^2} \implies 2x = 2y \implies x = y \implies [x]_{S^{2n-1}} = [y]_{S^{2m-1}}$. In either case, $H([x]_{S^{2m-1}}) = H([y]_{S^{2m-1}}) \implies [x]_{S^{2m-1}} = [y]_{S^{2m-1}}$, and H is injective.

Next consider any $(z, t) \in S^m$. If $(z, t) = (0, \dots, 0, -1)$, of course then it is the image of S^{m-1} by (*). Otherwise $t > -1$. Define

$$r = \sqrt{\frac{1-t}{2}}.$$

Since $(u, t) \in S^m$, we have $\|u\|^2 = 1 - t^2 = (1-t)(1+t)$. Let

$$x = \frac{u}{2\sqrt{1-r^2}}.$$

Then $\|x\| = r < 1$, and a direct substitution gives

$$h(x) = (u, t).$$

Hence H is surjective.

The map h is continuous, and $h = H \circ \pi$. Since π is a quotient map, H is continuous. Since B^m/S^{m-1} is compact and S^m is Hausdorff, the continuous bijection H is a homeomorphism.

Therefore

$$B^m/S^{m-1} \cong S^m.$$

Taking $m = 2n + 2$, we obtain

$$B^{2n+2}/S^{2n+1} \cong S^{2n+2}.$$

□

Problem 1. Prove that $\mathbb{R}P^1 \simeq S^1$. (One possible way to do this is to construct an explicit map and prove it is a homeomorphism.)

Proof. Here, we take the unit circle in $\mathbb{R}^2$ to be S^1 . In other words, we define $S^1 = \{\mathbf{x} \in \mathbb{R}^2 \mid \|\mathbf{x}\| = 1\}$ as a subspace of $\mathbb{R}^2$. Now, every line on $\mathbb{R}P^1$ intersects S^1 at two antipodal points, since S^1 is centered at the origin. So mapping each point of S^1 to the line it lies on through the origin would not be injective. This motivates our mapping.

Define $f : S^1 \rightarrow \mathbb{R}P^1$ via $f((\cos(\theta), \sin(\theta))) = [(\cos(\theta/2), \sin(\theta/2))] \in \mathbb{R}P^1$ where θ is the angle from $(1, 0)$ to each point on S^1 modulo 2π . Next, define $f^{-1} : \mathbb{R}P^1 \rightarrow S^1$ via $f^{-1}([\mathbf{x}]) = (\cos(2\theta_{[\mathbf{x}]}) , \sin(2\theta_{[\mathbf{x}]}))$ where $\theta_{[\mathbf{x}]}$ is the angle to the first antipodal point of $[\mathbf{x}]$ hit traveling counterclockwise from $(1, 0)$ to $[\mathbf{x}]$ along S^1 modulo π .

Clearly, each point on S^1 has a unique angle θ modulo 2π , so f is well-defined. Next, each $\theta_{[\mathbf{x}]}$ is clearly unique modulo π , which means each $2\theta_x$ is unique modulo 2π and so f^{-1} is well-defined. Observe.

$$\begin{aligned} f^{-1}(f(\cos(\theta), \sin(\theta))) &= f^{-1}([(\cos(\theta/2), \sin(\theta/2))]) = (\cos(\theta), \sin(\theta)) \\ f(f^{-1}([(\cos(\theta_{[\mathbf{x}]}, \sin(\theta_{[\mathbf{x}]}))])) &= f((\cos(2\theta_{[\mathbf{x}]}, \sin(2\theta_{[\mathbf{x}]}))) = [(\cos(\theta_{[\mathbf{x}]}, \sin(\theta_{[\mathbf{x}]})))] \end{aligned}$$

So we see f is a bijection. The fact that $\mathbb{R}P^1$ is compact is a result from class. So by **Theorem 5** We need only prove that f^{-1} is continuous, and that S^1 is Hausdorff to show that $S^1 \cong \mathbb{R}P^1$.

We begin with the latter. Pick any two distinct points $\mathbf{x}, \mathbf{y} \in S^1$. Since they are distinct, they have some positive distance $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| > 0$ via the Euclidean metric in $\mathbb{R}^2$. Let $r_{\mathbf{xy}} = d(\mathbf{x}, \mathbf{y})/2$. Then clearly the open balls $\mathcal{B}_{r_{\mathbf{xy}}}(\mathbf{x}), \mathcal{B}_{r_{\mathbf{xy}}}(\mathbf{y}) \subset \mathbb{R}^2$ are disjoint, and so are the sets $\mathcal{B}_{r_{\mathbf{xy}}}(\mathbf{x}) \cap S^1, \mathcal{B}_{r_{\mathbf{xy}}}(\mathbf{y}) \cap S^1$, which are open sets in the subspace S^1 by definition. So S^1 is Hausdorff.

Finally, since any open set of $\mathbb{R}^2$ is an arbitrary union of open balls, and any open set of S^1 is an intersection of an open set of $\mathbb{R}^2$ and S^1 , it is trivial that any open set of S^1 is just an arbitrary union of open arcs. Each an open arc in S^1 which we denote $(\mathbf{a} : \mathbf{b})$ via it's endpoints $\mathbf{a}, \mathbf{b} \in S^1$, has a unique angle range which defines it $(\theta_{\mathbf{a}}, \theta_{\mathbf{b}}) \subseteq [0, 2\pi)$. So $f^{-1}((\mathbf{a} : \mathbf{b}))$ is just all lines $[\mathbf{x}]$ with $\theta_{[\mathbf{x}]} \in (\theta_{\mathbf{a}}/2, \theta_{\mathbf{b}}/2)$. Well, just pullback these lines from $\mathbb{R}P^1$ to $\mathbb{R}^2 \setminus \{0\}$ and we get an open hourglass $H_{\mathbf{ab}}$ centered at the origin. Each point $\mathbf{x}$ has some minimal distances $\varepsilon_{\mathbf{a}}(\mathbf{x}), \varepsilon_{\mathbf{b}}(\mathbf{x})$, from the boundary rays through $\mathbf{a}$ and $\mathbf{b}$, respectively. (These distances are found by forming the lines perpendicular to both boundary rays through the point $\mathbf{x}$, they minimize the distance via *Pythagorean Theorem*). Set $r_{\mathbf{x}} = \min\{\varepsilon_{\mathbf{a}}(\mathbf{x}), \varepsilon_{\mathbf{b}}(\mathbf{x})\}/2$, and we can form an open ball in containing $\mathbf{x}$ completely within our hourglass $H_{\mathbf{a}, \mathbf{b}}$. Therefore, $\bigcup_{\mathbf{x} \in H_{\mathbf{ab}}} \mathcal{B}_{r_{\mathbf{x}}}(\mathbf{x}) = H_{\mathbf{ab}}$ and so $H_{\mathbf{ab}}$, the quotient preimage of $f^{-1}((\mathbf{a} : \mathbf{b}))$ is open in $\mathbb{R}^2 \setminus \{0\}$.

Therefore, by definition $f^{-1}((\mathbf{a} : \mathbf{b}))$ is open in $\mathbb{R}P^1$ and so the preimage of any arbitrary union of open arcs in S^1 via f^{-1} is open in $\mathbb{R}P^1$. So f^{-1} is a continuous bijection from $\mathbb{R}P^1$ to S^1 , and a homeomorphism by **Theorem 5**.

□

Problem 2. (a) Give n distinct homeomorphisms producing copies of $\mathbb{CP}^{n-1}$ in $\mathbb{CP}^n$.

(b) Let $q : S^{2n+1} \rightarrow \mathbb{CP}^n$ be the quotient map and $\phi : B^{2n} \rightarrow \mathbb{CP}^n$ be a map from the unit ball to complex projective space defined by $\phi(z) = q(z, \sqrt{1-|z|^2})$. Prove that $\phi(S^{2n-1}) = \mathbb{CP}^{n-1} \subset \mathbb{CP}^n$.

(c) Prove that $\mathbb{CP}^{n+1}/\mathbb{CP}^n \simeq S^{2n+2}$.

Proof. (a) In class we proved that $\mathbb{CP}^{n-1}$ is homeomorphic, say via $f : \mathbb{CP}^{n-1} \longleftrightarrow H_0$, to the projective hyperplane $H_0 = \{[0 : z_1 : \cdots : z_n] \mid (z_i)_{i \in [n]} \neq 0\}$ that lies in $\mathbb{CP}^n$. Well, of course all projective hyperplane subspaces in the collection $\{H_i = [z_0 : \cdots : z_i = 0 : \cdots : z_n] \text{ where } (z_j)_{j \in [n] \setminus \{i\}} \neq 0\}$ are homeomorphic to H_0 , say via $f_i : H_0 \longleftrightarrow H_i, \forall i \in [n]$. So $f_i \circ f, \forall i \in [n]$ are indeed the n distinct homeomorphisms desired, which make a total of $n+1$ homeomorphisms from $\mathbb{CP}^{n-1}$ to projective hyperplane subspaces H_i of $\mathbb{CP}^n$, which are the copies of $\mathbb{CP}^{n-1}$ in $\mathbb{CP}^n$.

(b) Take any $z \in S^{2n-1}$ which lies on the boundary of the unit ball $B^{2n} \subset \mathbb{C}^n$. By definition $|z| = 1$. So $\phi(z) = q(z, 0) = [z_0 : \cdots : z_{n-1} : 0] \in \mathbb{CP}^{n-1}$. Therefore, $\phi(S^{2n-1}) \subseteq H_n \subseteq \mathbb{CP}^n$. On the otherhand, take any $[z_0 : \cdots : z_{n-1} : 0] = [(z, 0)]$ in H_n . Well, $\frac{1}{|z|}(z, 0) \in [(z, 0)]$ and so $[(z, 0)] = [(\frac{1}{|z|}z, 0)] = q(\frac{1}{|z|}z, 0) = \phi(\frac{1}{|z|}z) \in \phi(S^{2n-1})$. Therefore, $H_n \subseteq \phi(S^{2n-1})$, and in fact $\phi(S^{2n-1}) = H_n$. Since $H_n \cong \mathbb{CP}^{n-1} \subseteq \mathbb{CP}^n$, ruffly $\phi(S^{2n-1}) = \mathbb{CP}^{n-1} \subseteq \mathbb{CP}^n$

(c)

We define $f : B^{2n+2}/S^{2n+1} \rightarrow \mathbb{CP}^{n+1}/H_{n+1}$ via $f([z]_{S^{2n+1}}) = [\phi(z)]_{H_{n+1}}$. We begin by showing f is well-defined. $[a]_{S^{2n+1}} = [b]_{S^{2n+1}} \implies a, b \in S^{2n+1} \text{ or } a = b \in B^{2n+2} \setminus S^{2n+1}$. If we have the former, by (b), $\phi(a), \phi(b) \in H_{n+1}$ and so $[\phi(a)]_{H_{n+1}} = [\phi(b)]_{H_{n+1}}$. Of course if the latter holds, $\phi(a) = \phi(b) \implies [\phi(a)]_{H_{n+1}} = [\phi(b)]_{H_{n+1}}$. So f is well-defined.

Now, suppose $f([x]_{S^{2n+1}}) = f([y]_{S^{2n+1}})$. If $x, y \in H_{n+1} = f(S^{2n+1})$, then of course their preimages belong to S^{2n+1} and therefore $[\phi^{-1}(x)]_{S^{2n+1}} = [y]_{S^{2n+1}}$. Otherwise, they do not belong to H_{n+1} and by definition $f([x]_{S^{2n+1}}) = f([y]_{S^{2n+1}}) \implies [x]_{S^{2n+1}} = [y]_{S^{2n+1}}$. So f is injective. Next, consider any $[x]_{H_{n+1}} \in \mathbb{CP}^{n+1}/H_{n+1}$. If $x \in H_{n+1} = \phi(S^{2n+1})$, then of course $\exists s \in S^{2n+1}$ such that $\phi(s) = x$ and so $f([s]_{S^{2n+1}}) = [\phi(s)]_{H_{n+1}} = [x]_{H_{n+1}}$. Otherwise, $x = [x_0 : \cdots : x_{n+1}] \in \mathbb{CP}^{n+1} \setminus H_{n+1}$, so $x_{n+1} \neq 0$. Well, $[x_0 : \cdots : x_{n+1}] = [\frac{1}{x_{n+1}}(x_0 : \cdots : x_{n+1})] = [\frac{x_0}{x_{n+1}} : \cdots : \frac{x_n}{x_{n+1}} : 1]$. let $u = \frac{1}{x_{n+1}}(x_0, \dots, x_n)$. Then $[u : 1] = [\frac{x_0}{x_{n+1}} : \cdots : \frac{x_n}{x_{n+1}} : 1]$ we solve for z such that $\phi(z, \sqrt{1-|z|^2}) = [u : 1]$.

A simple starting point is $[\frac{z}{\sqrt{1-|z|^2}} : 1]$, so $u = \frac{z}{\sqrt{1-|z|^2}} \implies |u| = \frac{|z|}{\sqrt{1-|z|^2}} \implies |u|^2 = \frac{|z|^2}{1-|z|^2} \implies |u|^2(1-|z|^2) = |z|^2 \implies |u|^2 = |z|^2(1+|u|^2) \implies |z|^2 = \frac{|u|^2}{1+|u|^2} \implies 1-|z|^2 = 1 - \frac{|u|^2}{1+|u|^2} = \frac{1}{1+|u|^2} \implies \sqrt{1-|z|^2} = \frac{1}{\sqrt{1+|u|^2}}$. Finally, $u = \frac{z}{\sqrt{1-|z|^2}} = \frac{z}{(\frac{1}{\sqrt{1+|u|^2}})} \implies z = \frac{u}{\sqrt{1+|u|^2}}$. So we have that $z = (\frac{u}{\sqrt{1+|u|^2}})$ is such a preimage in B^{2n+2} via ϕ and $f([z]_{S^{2n+1}}) = [u : 1]$ so f is surjective.

Let p, π be the quotient maps for B^{2n}/S^{2n+1} and $\mathbb{CP}^{n+2}/H_{n+1}$, respectively. Then $f \circ \pi = p \circ \phi$ by definition. Well, ϕ is defined by q , which is a quotient map and by definition continuous. Also p is continuous since it is a quotient map. So $p \circ \phi = f \circ \pi$ is continuous. Then since π is a quotient map and continuous, f must also be continuous. So f is a homeomorphism. So ruffly, $\mathbb{CP}^{n+1}/\mathbb{CP}^n \cong B^{2n+2}/S^{2n+1}$. Collapsing the boundary of a $2n+2$ -ball forms a S^{2n+2} sphere so we have the homeomorphism desired. $\square$

Problem 3. Prove that $S^n \wedge S^m \simeq S^{n+m}$ for any n and m .

Problem 4. Let X be a locally compact Hausdorff space. Given a point $x \in X$ and a neighborhood U of x , show that there is an open set V such that $x \in V \subseteq \overline{V} \subseteq U$ and $\overline{V}$ is compact.

Problem 5. Consider the smash product $X \wedge Y$ of two pointed spaces X and Y . Viewing this as a product, what is the unit E , i.e., for which topological space E is $X \wedge E \simeq X$? Now answer the same question for the wedge sum $X \vee Y$.

Problem 6. Consider $X = Y = \mathbb{Q}$ and $Z = \mathbb{N}$ with the subspace topology and with basepoint 0. Then $(X \wedge Y) \wedge Z \not\simeq X \wedge (Y \wedge Z)$. This shows that for pointed topological spaces, the smash product is not associative.

Your task (should you accept it, and you really should!) is to either

- (a) prove this, or
- (b) find a proof and write a brief sketch of it.

Remark: I know of only two “proofs” of this claim, first made in 1958 by Puppe. One of the “proofs” is at best a sketch; I do not think it is correct. The second was published in 2006 in a book by May–Sigurdsson (the actual counterexample being included in an appendix and having been worked out by Kathleen Lewis). This might serve as a hint as to which of the above you should try to do.